Manual of Elevator Load Weighing Controller DTZZIII-DC1S

V3.1.1.1.0

1 Technical Parameters and Type

Type	DTZZIII-DC1S	
Main Features	The analog output is 0~10mA current signal or 0~10V voltage signal Overload（NC）、Full_load（NO）、 Light_load（NO）	
Technical Parameters	Power supply	DC24（±10%）V
	Response time	≤0.5s
	Sensitivity	＜0.1 %
	Control return difference	＜1 %
	Working temperature	－10 ～ 60℃
	Power consumption of instrument	＜4W
	Overload capacity of sensor	150 %
	Output contactor capacity	1A/DC24V； 0.3A/AC220V
	Apply range	All kinds of movable bottom car elevator, installation clearance is about 15mm.

Notes:

1. **"NO" means normally open, when the relay is activated, the contact will be closed.**
2. **"NC" means normally closed, when the relay is activated, the contact will be open.**

2 Controller Appearance and Terminal Functions

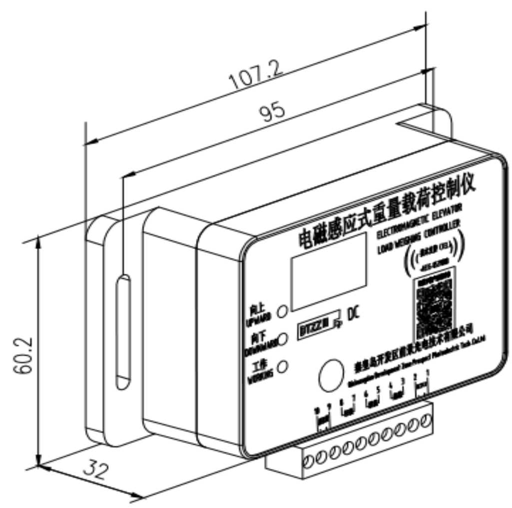

Figure 1 Outline diagram of controller

2.1 Indicator light：

Upward indicator light: This light is bright when clearance is more than or equal to 17mm. Controller should be adjusted upwards at this time.

Downward indicator light: This light is bright when clearance is more than or equal to 0 mm and less than or equal to 14mm. Controller should be adjusted downwards at this time.

Working indicator light: This light is bright when working normally. In self-learning, it is used to indicate whether the learning is correct or not.

2.2 Controller Terminal Functions：

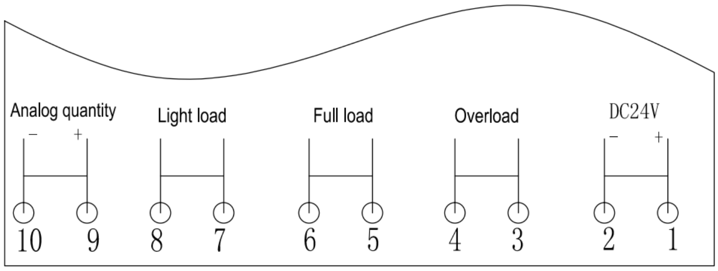

Figure 2 Controller terminal schematic diagram

2.2.1 Terminals 1 and 2 are controller DC24V power supply. Terminal 1 is positive, and terminal 2 is negative.

2.2.2 Terminals 3 and 4 are contactor of overload relay. （DC：NC；DC-2：NO）

2.2.3 Terminals 5 and 6 are contactor of full-load relay. （NO）

2.2.4 Terminals 7 and 8 are contactor of light-load relay. （NO）

2.2.5 Terminals 9 and 10 are 0~10mA current signal or 0~10V voltage signal output by the instrument, 9 is positive and 10 is negative. If the controller is too far away from the inverter, add a 1K resistor to the signal input terminal of the inverter as shown in the figure below.

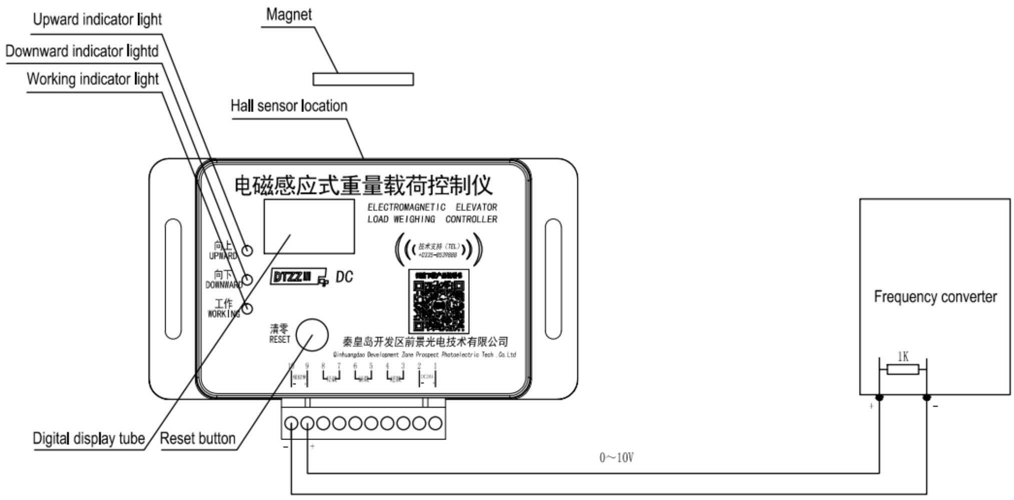

Figure 3 Schematic diagram of long-distance transmission of compensation signal

3 Controller Size and Installation Method

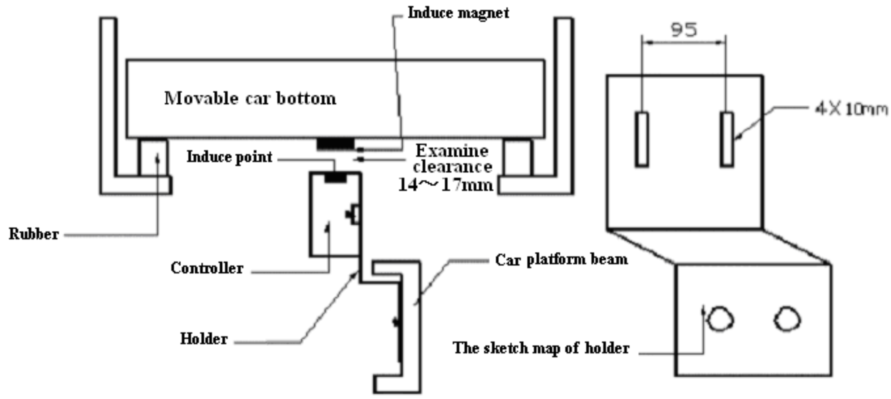

Figure 4 Controller installation schematic diagram

3.1 When the elevator car is empty, magnet is adsorbed under the car bottom center position and the symbol side is downward.

3.2 Use M3 bolts to install the controller on the fixed bracket, and then fix the fixed bracket on the bottom beam of the car at the position corresponding to the magnet, so that the middle position of the induction indicated on the controller is directly opposite to the middle position of the induction magnet, to ensure that the mark on the magnet is strictly aligned with the mark on the controller face to face, and the controller is parallel to the magnet.

3.3 To make the clearance about 15mm by adjusting controller, one of the two indicator lights is bright if installation position is not right. Adjust controller upwards if the upward indicator light is bright. Adjust controller downwards if the downward indicator light is bright. The two indicator lights are all off if controller position is right.

3.4 The examine clearance can be added to above 15mm if car platform displacement range exceed 15mm in practice installation. But the output signal of nearby empty load will have excursion. So this kind of installation method isn’t suggested.

Notes :

1) The fixed bracket of the controller needs to be made by the user according to the specific situation.

2) The magnet used in the controller is a special circle powerful magnet.

3) It is recommended to choose a rubber block whose car bottom displacement is about 6-8mm from no-load to rated load.

4 Production Debugging

4.1 Install the controller on the bottom of the car, turn on the power without load, and the three-digit digital tube will display "821",then "888" is displayed, at this time, you can observe whether the digital tube is missing;

4.2 Clear the elevator without load. Press the Reset button three times, the working indicator (Working) goes out, accompanied by three soft sounds, about 5 seconds or so, three long responses are heard, and the working indicator (Working) lights up again. Reset operation over, the digital tube displays "000";

4.3 Add loading-weight to light-load value, press the “Reset” button 5 times continuously in 5 seconds. The working indicator light of controller goes out with 5 short sounds at the same time. About 5 seconds later, you will hear one long answering sounds, and the working light of controller will be bright again. It means that the study of light-load point has been finished (If you don't need this point, you can skip this step).

4.4 Add loading-weight to rated load value, keep pressing the "Reset" button more than 5 seconds. The working indicator light of controller goes out with one short sounds at the same time. The working light of controller will be bright again when you hear one short answering sound and one long answering sound. It means that the study of full-load point has been finished, then the 3 digital tubes of controller will display "100".

5 Controller Introductions

5.1 When the self-learning operation is performed, the working indicator is off, and the working indicator lights up again when the operation is completed. The next learning operation can only be performed when the working indicator is on.

5.2 During self-learning, when the working indicator goes from off to on, if there is no correct echo of the corresponding operation, it means that the operation has failed and the operation needs to be re-learned.

5.3 The downward indicator light and working indicator light are bright when controller works normally after self-learning.

Debugging is over and controller can work normally. When the load weight of the car changes from no-load to rated load, the instrument outputs the corresponding 0~10mA (the external supporting 1kΩ resistor can output 0~10V), the digital tube display changes from "000" to "102" or more. Light-load relay active when loading weight in car reach the preset light-load value. Full-load relay active when loading weight in car reach 95% rated load. Overload relay active when loading weight in car reach 102% rated load.

If there is any questions in the scene or there is something wrong with the controller, please get in touch with the technical staffs of our company in time.

Technical support tel: 0335-8539888/8539818

Note: When the product is scrapped, the relevant disposal regulations related to the list of hazardous wastes must be strictly implemented.

Manufacturer: Qinhuangdao Development Zone Prospect Photoelectric Tech. Co., Ltd.

Address: No. 3 Jinghe road, Huashan zhonglu M, Qinhuangdao Economic & Technological Development Zone,Hebei Province.

Post Code: 066004